



FOUR-COURSE SYSTEM

COURSE 3 - DATA ENGINEERING 2026

C3-043 RapidRoute Orchestration Mini-Lab

Fresh limited-hint integration after all 13 lesson mastery records.

Artifact	C3-043
Status	2026 - LEARNER EDITION
Phase	C3-03 Orchestration & Reliable Workflows 13 lessons 34-46 focused hours
Prerequisite	C3-02 Gate PASS; C2B Python/quality/ownership foundations assumed, orchestration taught here.
Brand	Charcoal #1F2937 Teal #14B8A6 Soft Blue #3B82F6 AMOLED #000

House teaching rule: simple mental picture -> tiny example -> guided Airflow/tool workflow -> break/fix -> independent changed task -> validation -> production reasoning -> evidence. A green scheduler box alone never proves correct data.



RapidRoute scenario

RapidRoute receives one shipment-event file per service date. Operations needs a reliable daily current-state shipment pipeline that waits for the partition, validates it, loads/merges safely, reconciles, publishes by interval and can reprocess history without regressing newer state.

Architecture picture

scheduled DagRun -> resolve interval -> wait/readiness -> extract -> validate -> idempotent load/merge -> reconcile -> atomic publish -> operator evidence

Setup

Unzip C3_043_RAPIDROUTE_MINI_LAB_STARTER_RC1.zip into a fresh learner workspace. Do not reuse your practice output database/state.

Staged build

- State one-run interval and business side effect before coding.
- Complete the fresh RapidRoute TaskFlow DAG; keep pure pipeline functions outside the DAG wrapper.
- Use interval-derived Oct 1-3 shipment paths.
- Add bounded readiness waiting, retry/timeout policy and a named sink pool.
- Run pure tests/static checks before scheduler execution.
- In WSL2 Airflow, discover the DAG and run the supplied dates.
- Repeat one logical date and prove convergence; replay an older date after newer state exists.
- Inject the malformed file and prove deterministic bad data remains failed.
- Write detect -> diagnose -> recover -> validate runbook evidence.

Required competencies

- DAG architecture and meaningful task boundaries
- Interval correctness and historical partition mapping
- Failure classification, bounded retry/timeout
- Idempotent same-date rerun and stale-safe historical reprocessing
- Readiness wait and scarce-resource controls
- Testing/discovery/troubleshooting evidence
- Operator runbook and professional handoff

Pass rule

Every required competency above must PASS and there must be zero unresolved critical failure. A missing scheduler/runtime proof cannot be averaged away.

Critical failures

- Wrong interval/wall-clock partitioning corrupts historical replay.
- Same logical run duplicates/regresses business state.
- Validation can fail while publish still succeeds.
- Secret/credential is committed or exposed in evidence.
- Deterministic bad data is retried into a false success.